

ADS2 Mock Coding Challenge 2

Semester 2, 2024-25

Technical Instructions

You have 3 hours to complete this assignment. There are **three** questions, all of which need to be completed. The instructions and data sets (`.csv` or `.txt` files) can be downloaded from Blackboard Learn.

The final submission is a PDF knitted from the R Markdown file (if you cannot knit to PDF directly, then knit to Word and convert the outcome to a PDF file using the “Export” function in Word or another text editor). A template is provided. The submission should contain explanatory text, answers to questions, all results, and all the code used to generate the results. There is one exception: When you read a `.csv` file and if your name is in the file path, you are allowed to hide that code chunk so that your anonymity is maintained.

You will be graded not only on your answers to the questions, but also on your ability to compile a well-formatted and readable R Markdown document. It is therefore advisable to knit early and often, and check that your document can be knitted without errors and that the result is in line with your expectations. If you have code chunks that take a long time to run, use the code chunk option `cache = TRUE`. This means that the results of the code chunk get saved and will be used in the next knit, instead of being computed again (provided the code chunk has not changed).

Please upload your PDF file to the assessment dropbox at the end of the assignment. We are aware that due to increased traffic when everybody uploads their file, your upload may be a few minutes past the deadline. In such cases, we will consult the time at which the PDF document was produced and use this to determine whether or not your submission counts as a late submission. If so, the same penalties apply as for other in-course assessments.

Honour Code

This is an open-book assessment. This means you are allowed to consult your previous notes and use your previous code. You are also allowed to look up commands online if you need to (though the assessment is designed in such a way that you should not need commands or methods beyond what has been taught in this class). If you use code from an online source, please state what the source is (name of site, author if possible, url, date accessed).

You are **not** allowed to work with other students or use AI copilots (even if they are from the allowed websites). Of course, because we are allowing internet access, we cannot completely rule out the possibility of you working together. We appeal to your sense of honour and integrity. Also, be aware that the activity on your computers ***will be monitored***.

By submitting this assignment, you declare that this is the result of your own work and that you did not get help from other students or AI copilots or helped others. If, in marking the finished work, we find evidence that students have colluded, this will be treated as a potential violation of academic integrity and brought before the ZAMO. For more details, refer to the academic misconduct section of the course handbook.

Treatment for type 1 diabetes (T1D)

Your team developed an insulinomimetic that can be used to substitute insulin in patients with T1D. You induced T1D in mice by injecting them with streptozotocin (this drug destroys pancreatic β -cells, which abrogates insulin production and leads to the loss of control over glucose metabolism). Subsequently, mice were injected with vehicle or this novel drug in 2 possible concentrations. Their blood glucose level (mM) was measured before the injection and 1 hour after.

Answer the questions below and provide clear and reproducible code as well as your comments.

Questions

- Import, check, and organize the data appropriately. Treat data and reformat columns if needed. (3 points.)
- Plot the data in a useful way. (5 points.)
- Choose, justify, state the statistical hypotheses, and carry out an appropriate test to answer whether the drug is useful. (13 points.)
- Present and discuss your results. Is this novel drug useful? What would you suggest doing next? (5 points.)